

Task 3

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Term	Full Form	One-line Use
HTTP	HyperText Transfer Protocol	Used to transfer web pages between a browser and a web server.
HTTPS	HyperText Transfer Protocol Secure	Used to securely transfer data between a browser and a web server using encryption.
FTP	File Transfer Protocol	Used to upload and download files between computers over a network.
SMTP	Simple Mail Transfer Protocol	Used to send email from one mail server to another.

2. What is the difference between IPv4 and IPv6? (2 lines)

- **IPv4** uses **32-bit** addresses (e.g., **192.168.1.1**) and supports about **4.3 billion** unique addresses.
- **IPv6** uses **128-bit** addresses (e.g., **2001:db8::1**) and supports a vastly larger number of unique addresses.

3. Run `nslookup google.com` on your system and write down the IP address you get.

I can't run commands on your computer, so I can't see the result directly.

Run this command in **Command Prompt** (Windows) or **Terminal** (Linux/macOS):

```
1 | nslookup google.com
```

Example output (your result may be different depending on your location and DNS server):

```
1 | Name:      google.com
```

Note: Google uses multiple IP addresses, so your output may show a different valid IP address.

4. What do these HTTP status codes mean?

Status Code	Meaning
200 OK	The request was successful.
301 Moved Permanently	The resource has been permanently moved to a new URL.
403 Forbidden	The server understood the request but refuses access.
404 Not Found	The requested page or resource could not be found.
500 Internal Server Error	The server encountered an unexpected error while processing the request.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with as an example.

For `www.google.com` :

- **Root (.)** → The highest level of the DNS hierarchy (usually hidden).
- **TLD (Top-Level Domain)** → `.com`
- **Domain Name** → `google`
- **Subdomain** → `www`

Hierarchy:

```
1 Root (.)
2   |
3   .com   ← TLD
4   |
5   google ← Domain
6   |
7   www    ← Subdomain
```

When you type  into your browser, the DNS system follows this hierarchy to find the correct IP address of Google's server.